

Q1. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

There are 48 coins on a table in a single group. The coins are first divided into two separate groups such that the number of coins in each group is an even number. The product of the number of coins in the two groups is noted and the coins in each group are again divided (if possible) into two groups, each containing an even number of coins, and again the product of the numbers in each of the two smaller groups is noted. This process continues until there are 24 groups containing two coins each. What is the sum of all the products?

- ☐ a) **1012**
- ☐ b) **1104**
- ☐ c) **506**
- ☐ d) **Depends on the way the coins are divided**

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.

An instruction was sent to all the railway stations across the country that, workers above the age of 47 years must retire. Five workers with different ages at the Kazipet railway station managed to tamper their records. However, the station master knew that the sums of the ages (in years) of all the possible pairs of workers (from out of the five) are 102, 105, 107, 107, 109, 109, 111, 112, 114, 116. Find the age of the oldest of the five workers.

- ☐ a) 60 years
- ☐ b) 58 years
- ☐ c) 59 years
- ☐ d) Cannot be determined

Q3. DIRECTIONS for questions 1 to 5: Select the correct alternative from the given choices.

There are three containers – M, W and H – of equal capacities, containing milk, water and honey respectively. Each container is filled up to 75% of its capacity. Now, W is topped up to its brim (i.e., filled completely), with the contents of M. The contents of W are then thoroughly mixed. Then H is topped up to its brim with the contents of W. The contents of H are then thoroughly mixed. Finally, the contents of H are used to fill M again up to 75% of its capacity. Which among the following statements is true regarding m , w and h , which are the respective final concentrations of milk in M, water in W and honey in H?

☐ a) $m > w > h$

☐ b) $w > h > m$

☐ c) $m = w = h$

☐ d) $m < w = h$

Q4. DIRECTIONS for questions 1 to 5: Select the correct alternative from the given choices.

a , b , c and p are distinct positive integers, each greater than 1, and c is a prime number. Which of the following statements is/are necessarily true, if $ab + c$ is exactly divisible by p ?

I. Either a or b is divisible by p .

II. Neither a nor b is divisible by p .

III. ab when divided by p gives a remainder of c .

IV. Sum of the remainders of the divisions ab/p and c/p add upto p .

☐ a) Only I and IV

☐ b) Only I and III

☐ c) Only II and IV

☐ d) Only II and III

Q5. DIRECTIONS for questions 1 to 5: Select the correct alternative from the given choices.

Mr. Ram sold an article at a certain profit. Had he purchased the article for 12% less than what he actually purchased it for, and then sold it for 24% more than what he actually sold it for, his profit percentage would have been $2\frac{1}{2}$ times his actual profit percentage. Find his actual profit percentage.

☐ a) $27\frac{1}{2}\%$

☐ b) $28\frac{2}{7}\%$

☐ c) $33\frac{1}{3}\%$

☐ d) $37\frac{1}{2}\%$

Q6. DIRECTIONS *for question 6:* Type in your answer in the input box provided below the question.

Amar, Bhavan and Chetan have a total of 40 marbles among them. First, Amar tripled the number of marbles with each of the other two by giving them some marbles. Next, Bhavan similarly triples the number of marbles with each of the other two. If Bhavan finally has 10 marbles, find the number of marbles he had initially.

Q7. DIRECTIONS *for question 7:* Select the correct alternative from the given choices.

A group of 24 workers, comprising 8 men, 8 women and 8 boys is assigned a piece of work. No two workers in the group have the same capacity to work. Each day, a sub-group of workers, comprising 1 man, 1 woman and 1 boy, work together. In this manner, sub-groups of all possible combinations comprising one man, one woman and one child work on exactly one day each, and the work is completed on the last day, taking a total of 512 days. If all the 24 workers work together for one day, what part of the entire work will be completed?

- ☐ a) $1/8^{\text{th}}$
- ☐ b) $1/64^{\text{th}}$
- ☐ c) $1/32^{\text{nd}}$
- ☐ d) $1/16^{\text{th}}$

Q8. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

The number of positive integral ordered pairs (x, y) , that satisfy the equation $3x + xy = 138 - 2y$ is

Q9. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

In a horticultural show, a participant arranged a total of 1000 fruits, of four types, in a long straight line. If he first placed one apple, then two mangoes, three oranges, four custard apples, five apples, six mangoes, seven oranges, eight custard apples and so on (with apples followed by mangoes followed by oranges followed by custard apples) up to the 1000th fruit, what was the position of the 100th mango?

Q10. DIRECTIONS *for question 10:* Select the correct alternative from the given choices.

The lengths (in cm) of the sides of a 100-sided convex polygon are 1, 2, 3100. A circle of radius r cm is inscribed in this polygon. What is the area of the polygon?

- ☐ a) $5050r \text{ cm}^2$
- ☐ b) $2525r \text{ cm}^2$
- ☐ c) $5005r \text{ cm}^2$
- ☐ d) $5555r \text{ cm}^2$

Q11. DIRECTIONS *for question 11:* Type in your answer in the input box provided below the question.

In how many ways can three married couples sit around a circular table such that no pair of wife and husband are opposite each other?

Q12. DIRECTIONS *for questions 12 to 19:* Select the correct alternative from the given choices.

If $a = \sqrt[2]{b} = \sqrt[3]{c} = \sqrt[4]{d} = \sqrt[5]{e} = \sqrt[2]{6}$, find the number of digits in the number $(a \times b \times c \times d \times e)^{20}$, given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$.

- ☐ a) **116**
- ☐ b) **117**
- ☐ c) **118**
- ☐ d) **115**

Q13. DIRECTIONS *for questions 12 to 19:* Select the correct alternative from the given choices.

When a number N is divided by D the remainder is 23 and when N is divided by $12D$ the remainder is 104. What is the remainder when N is divided by $6D$?

☐ a) 104

☐ b) 23

☐ c) 27

☐ d) 81

Q14. DIRECTIONS for questions 12 to 19: Select the correct alternative from the given choices.

If $a^{(p-q)} \cdot b^{(q-r)} \cdot c^{(r-p)} = 1$, and a, b, c, p, q and r are rational numbers, then which of the following statements is necessarily true?

☐ a) a, b and c must all be equal.

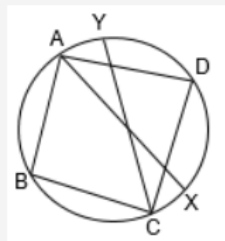
☐ b) $p = q = r$

☐ c) **Either (A) or (B)**

☐ d) **None of the above**

Q15. DIRECTIONS for questions 12 to 19: Select the correct alternative from the given choices.

In the figure below, ABCD is a cyclic quadrilateral in which the angle bisector of $\angle BAD$ meets the circle in X, while the angle bisector of $\angle BCD$ meets the circle in Y. If $XY = 12$ cm, find the area of the circle.



- ☐ a) 36π sq.cm
- ☐ b) 72π sq.cm
- ☐ c) 60π sq.cm
- ☐ d) 49π sq.cm

Q16. DIRECTIONS *for questions 12 to 19:* Select the correct alternative from the given choices.

A function $f(x)$ is defined for all real values of x , such that $f(x + y) = \frac{f(x) + f(y)}{1 - f(x)f(y)}$, where $f(x) \neq \frac{1}{f(y)}$. If $f(n) > 0$ and $f(2n) = 1$, what is the value of $f(3n)$?

☐ a)

$\sqrt{2} - 1$

☐ b) 1

☐ c) $\frac{3}{2}$

☐ d)

$\sqrt{2} + 1$

Q17. DIRECTIONS *for questions 12 to 19:* Select the correct alternative from the given choices.

Let x, y, z be three positive numbers, such that $x + y + z = 1$. The minimum possible value of $\left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2 + \left(z + \frac{1}{z}\right)^2$ is

- ☐ a) **27.**
- ☐ b) **18.**
- ☐ c) **$33\frac{1}{3}$**
- ☐ d) **40.**

Q18. DIRECTIONS *for questions 12 to 19:* Select the correct alternative from the given choices.

How many circles exist in the x - y plane such that they touch both the coordinate axes and pass through the point $(3, 6)$?

☐ a) 0

☐ b) 1

☐ c) 2

☐ d) 3

Q19. DIRECTIONS *for questions 12 to 19:* Select the correct alternative from the given choices.

In a relay race, each team member runs only a certain (not necessarily equal) portion of the total length of the race, such that together all the team members cover the entire length of the race. A certain relay race is conducted, with each team comprising exactly two runners. A and B are two runners of one of the teams participating in the relay race. The total length of the race is 140 m and the total time that A and B take to finish the race is 20 seconds. If it turns out that the average speed of A and B together for the entire race equals the average of their individual speeds, how many of the following statements are false?

- I. If A's speed is 6 m/s, B's speed is 8 m/s.
- II. If B's speed is more than 8 m/s, A runs for less than 12 seconds.
- III. If B's speed is 7 m/s, A's speed is 7 m/s.
- IV. If A's speed is less than 6 m/s, B runs for more than 12 seconds.

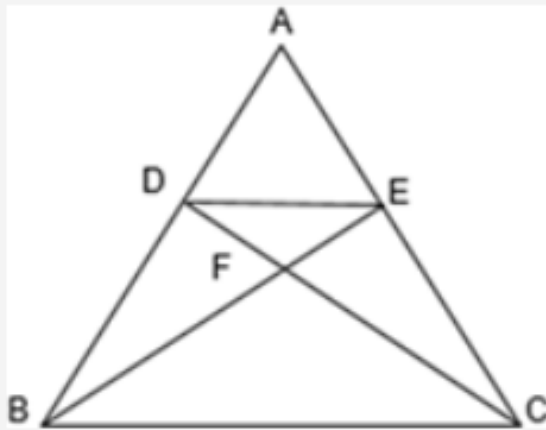
- ☐ a) 0
- ☐ b) 1
- ☐ c) 2
- ☐ d) **More than 2**

Q20. DIRECTIONS *for questions 20 to 22:* Type in your answer in the input box provided below the question.

Nine boxes are numbered 1, 2, 9. Each box must be filled with a ball which is of one of the three colours white, black and yellow. All the boxes which are filled with either a white ball or a black ball must have consecutive numbers. At least 6 boxes must be filled with a yellow ball. Find the number of ways in which the boxes can be filled.

Q21. DIRECTIONS *for questions 20 to 22:* Type in your answer in the input box provided below the question.

In the triangle ABC shown below, DE is parallel to BC and D divides AB in the ratio 1 : 3.



If the area of triangle ADE is 5 sq.cm, what is the area of the triangle EFC (in sq.cm)?

Q22. DIRECTIONS *for questions 20 to 22:* Type in your answer in the input box provided below the question.

There is a compound wall along the perimeter of a rectangular plot. Two walls of the plot run North-South and two walls run East-West. A man is standing x feet to the north of the center of the north wall. A second man starts walking eastwards from the South-East corner of the compound and can just see the first man after walking a distance of exactly 100 feet. If he walks another 200 feet further to the East, the line joining the two men makes an angle of $\tan^{-1} \frac{3}{4}$ with the south wall, while the line joining the North-East corner of the compound to the second man makes an angle of $\tan^{-1} \frac{1}{2}$ with the south wall. Find x .